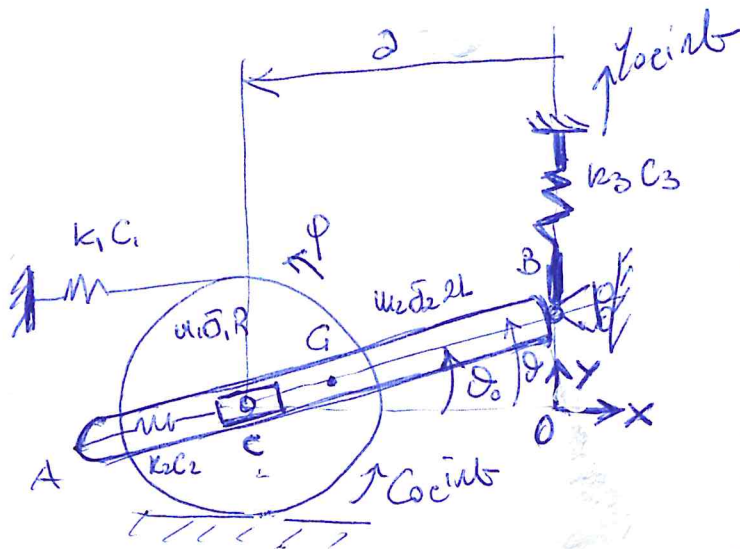




$$\frac{d\bar{y}_G}{d\theta} = \begin{cases} 1/\cos^2\theta \\ 1+\bar{y}_G^2\theta \end{cases}$$

$$\frac{d}{d\theta}\left(\frac{1}{\cos\theta}\right) = \frac{\sec\theta}{\cos^2\theta}$$

$$x = \begin{Bmatrix} 0 \\ \varphi \end{Bmatrix}$$

$$x_c = -a - R\varphi$$

$$y_B = (a + R\varphi)\bar{y}_G\theta$$

$$x_q = -L\cos\theta$$

$$y_q = (a + R\varphi)\bar{y}_G\theta - L\sec\theta$$

$$\bar{c}_B = (a + R\varphi)/\cos\theta$$

$$\Delta L_1 = -2R\varphi$$

$$\Delta L_2 = \bar{AC} - (\bar{AC})_0 = 2L - \frac{(a + R\varphi)}{\cos\theta} - \left(2L - \frac{a}{\cos\theta_0}\right) =$$

$$= -\frac{a + R\varphi}{\cos\theta} + \frac{a}{\cos\theta_0}$$

$$\Delta L_3 = -(y_B - y_{B0}) = y_{B\varphi} - (a + R\varphi)\bar{y}_G\theta$$

$$[u_{ph}] = \text{diag}(m_1 \bar{\vartheta}, m_2 \bar{\vartheta}_2) \quad , \quad \underline{F} = \{ \bar{c}_B \bar{c}_1 \bar{c}_2 \}^T$$

$$\frac{\partial x_c}{\partial \theta} = 0 \quad , \quad \frac{\partial x_c}{\partial \varphi} = -R$$

$$\frac{\partial x_q}{\partial \theta} = L\sec\theta \quad , \quad \frac{\partial x_q}{\partial \varphi} = 0$$

$$\frac{\partial y_q}{\partial \theta} = (a + R\varphi) \frac{1}{\cos^2\theta} - L\cos\theta \quad , \quad \frac{\partial y_q}{\partial \varphi} = R\bar{y}_G\theta$$

$$\frac{\partial \Delta L_1}{\partial \theta} = 0 \quad , \quad \frac{\partial \Delta L_1}{\partial \varphi} = -2R$$

$$\frac{\partial \Delta L_2}{\partial \theta} = -(a + R\varphi) \frac{\sec\theta}{\cos^2\theta} \quad , \quad \frac{\partial \Delta L_2}{\partial \varphi} = -\frac{R}{\cos\theta}$$

$$\frac{\partial \Delta L_3}{\partial \theta} = -(a + R\varphi) \frac{1}{\cos^2\theta} \quad , \quad \frac{\partial \Delta L_3}{\partial \varphi} = -R\bar{y}_G\theta$$

$$\frac{\partial y_B}{\partial \theta} = (a + R\varphi) \frac{1}{\cos^2\theta} \quad , \quad \frac{\partial y_B}{\partial \varphi} = R\bar{y}_G\theta$$

$$\Delta u = \begin{bmatrix} 0 & -R \\ 0 & 1 \\ L \sin \theta_0 & 0 \\ \frac{2}{\cos^2 \theta_0} - L \cos \theta_0 & R \tan \theta_0 \\ 1 & 0 \end{bmatrix} \begin{matrix} x_c \\ \varphi \\ x_q \\ x_q \\ \theta \end{matrix}$$

$$\Delta u = \begin{bmatrix} 0 & -2R \\ -\frac{2 \sin \theta_0}{\cos^2 \theta_0} & -\frac{R}{\cos \theta_0} \\ -\frac{2}{\cos^2 \theta_0} & -R \tan \theta_0 \end{bmatrix} \begin{matrix} \Delta u_1 \\ \Delta u_2 \\ \Delta u_3 \end{matrix}$$

$$\Delta u = \begin{bmatrix} 1 & 0 \\ \frac{2}{\cos^2 \theta_0} & R \tan \theta_0 \end{bmatrix} \begin{matrix} \text{Comb} \\ \text{Tel-y} \end{matrix}$$

$$\frac{\partial^2 \varphi_q}{\partial \theta^2} = (\partial + R \varphi) \frac{2 \cos \theta \sin \theta}{\cos^3 \theta} + L \sin \theta$$

$$\frac{\partial \varphi_q}{\partial \theta \partial \varphi} = \frac{R}{\cos^2 \theta}$$

$$\frac{\partial \varphi_q}{\partial \varphi^2} = 0$$

$$[K_{q2}] = u_2 g \begin{bmatrix} \frac{2 \sin \theta_0}{\cos^3 \theta_0} + L \sin \theta_0 & \frac{R}{\cos^2 \theta_0} \\ \frac{R}{\cos^2 \theta_0} & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = -(\partial + R \varphi) \frac{\cos^3 \theta + 2 \sin^2 \theta \cos \theta}{\cos^4 \theta}$$

$$\frac{\partial \Delta L_2}{\partial \theta \partial \varphi} = -\frac{R \sin \theta}{\cos^2 \theta}$$

$$\frac{\partial \Delta L_2}{\partial \varphi^2} = 0$$

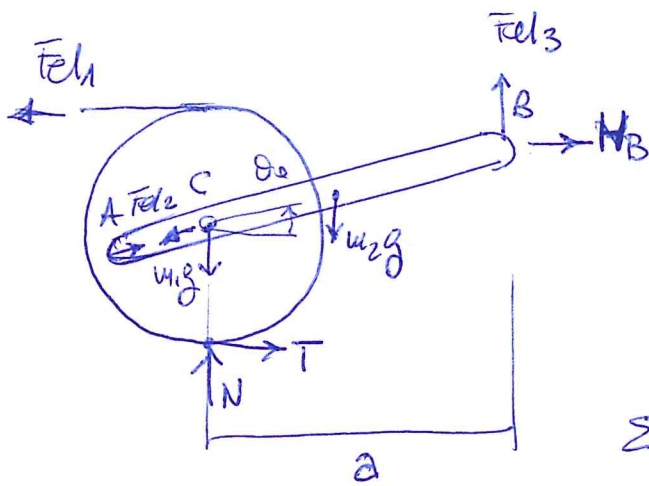
$$[K_{EL II 2}] = K_2 \Delta L_2 \begin{bmatrix} -\frac{2 \sin^2 \theta_0}{\cos^3 \theta_0} & \frac{R \sin \theta_0}{\cos^2 \theta_0} \\ -\frac{R \sin \theta_0}{\cos^2 \theta_0} & 0 \end{bmatrix}$$

$$\frac{\partial \Delta L_3}{\partial \theta^2} = -(\partial + R \varphi) \frac{2 \cos \theta \sin \theta}{\cos^3 \theta}$$

$$\frac{\partial \Delta L_3}{\partial \theta \partial \varphi} = -\frac{R}{\cos^2 \theta}$$

$$\frac{\partial \Delta L_3}{\partial \varphi^2} = 0$$

$$[K_{EL II 3}] = K_3 \Delta L_3 \begin{bmatrix} -\frac{2 \sin \theta_0}{\cos^3 \theta_0} & -\frac{R}{\cos^2 \theta_0} \\ -\frac{R}{\cos^2 \theta_0} & 0 \end{bmatrix}$$



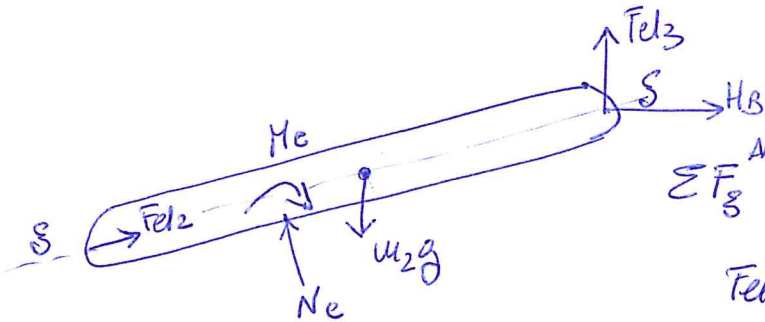
$$\sum M_C^{DISCO} = 0 \rightarrow \text{~~F_{el1} L \cos \theta_0~~}$$

$$T = -F_{el1}$$

$$\sum F_x^{TOT} = 0 \rightarrow H_B = 2F_{el1}$$

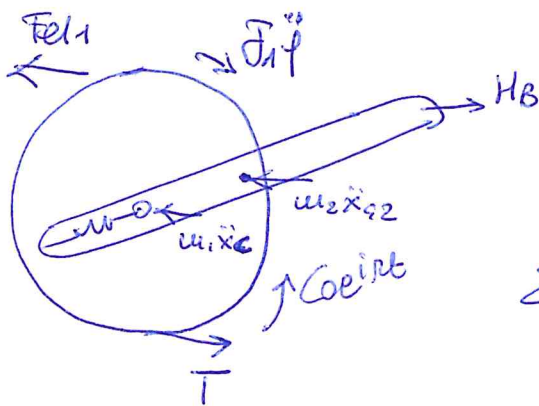
$$\sum M_C^{AB} = 0 \rightarrow F_{el3} L - u_2 g (L - L \cos \theta_0) = 0$$

$$F_{el3} = u_2 g \frac{L - L \cos \theta_0}{L} + \frac{H_B L \sin \theta_0}{2F_{el1} L \sin \theta_0}$$



$$\sum F_s^{AB} = 0: F_{el2} - u_2 g \sin \theta_0 + F_{el3} \sin \theta_0 + H_B \cos \theta_0 = 0$$

$$F_{el2} = (u_2 g - F_{el3}) \sin \theta_0 - 2F_{el1} \cos \theta_0$$



$$\sum M_C^{DISCO} = 0: \text{~~F_{el1} L \cos \theta_0~~}$$

$$F_{el1} R + T R - J_1 \ddot{\varphi} R = 0$$

$$T = -F_{el1} + \frac{J_1 \ddot{\varphi}}{R} - \frac{Coe^{iRt}}{R}$$

$$\sum F_x^{TOT} = 0: H_B = F_{el1} - T + u_1 \ddot{x}_C + u_2 \ddot{x}_{S2}$$

